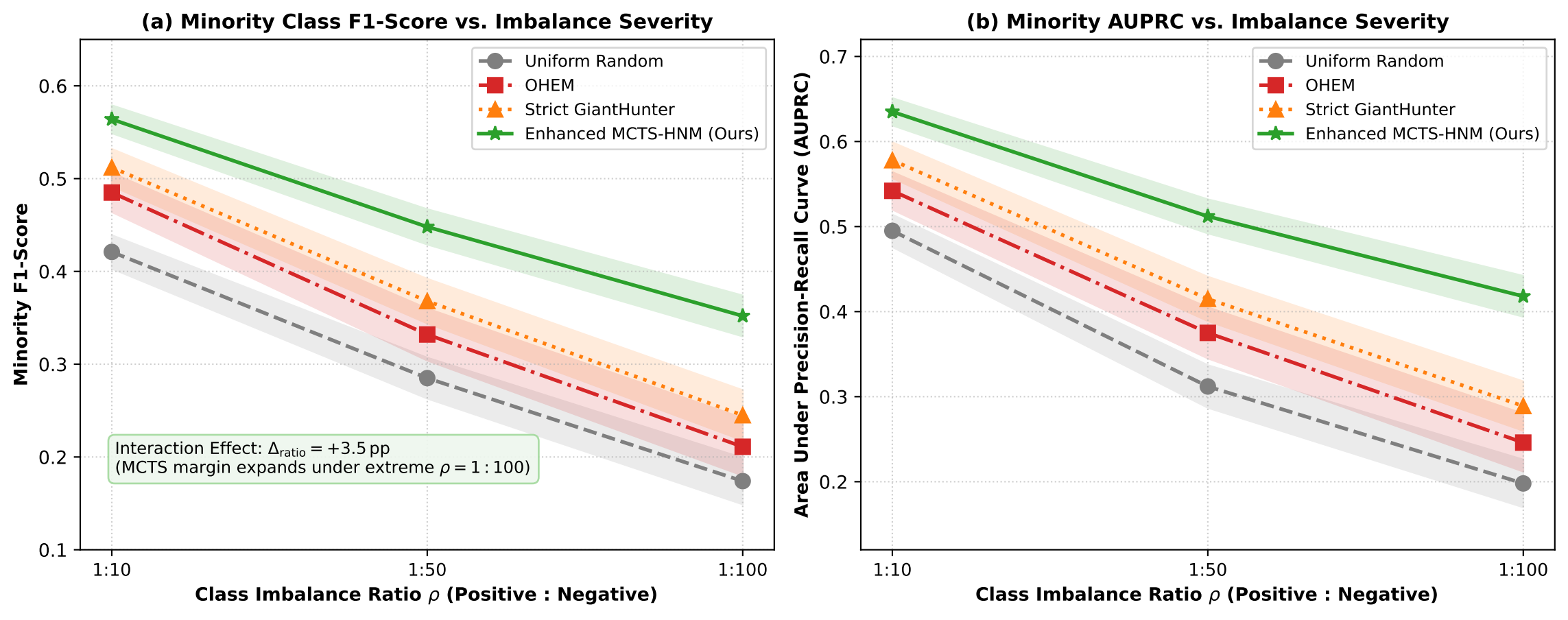
(a) Minority Class F1-Score vs. Imbalance Severity
Minority F1-Score
Class Imbalance Ratio ρ (Positive : Negative)
1:10
1:50
1:100
0.1
0.2
0.3
0.4
0.5
0.6
Uniform Random
OHEM
Strict GiantHunter
Enhanced MCTS-HNM (Ours)
Interaction Effect: Δratio = +3.5 pp
(MCTS margin expands under extreme ρ = 1 : 100)
(b) Minority AUPRC vs. Imbalance Severity
Area Under Precision-Recall Curve (AUPRC)
Class Imbalance Ratio ρ (Positive : Negative)
1:10
1:50
1:100
0.2
0.3
0.4
0.5
0.6
0.7
Uniform Random
OHEM
Strict GiantHunter
Enhanced MCTS-HNM (Ours)